

SPRiF: Spectral Projective Reset Integrate-and-Fire Neuron

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Abstract

Temporal tasks often require information to persist across long intervals. Standard spiking neurons use the same membrane state to accumulate context, generate spikes, and receive post-spike reset. This coupling repeatedly perturbs the temporal trace that a neuron must preserve. We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), a spiking unit based on *functional state decomposition*. SPRiF separates a constrained spectral slow state, which provides real and rotational memory modes without reset, from a fast discharge state that generates spikes and receives a learnable projective reset. The design preserves a resettable spike-generation pathway while insulating slow temporal memory from post-spike correction. Across five temporal classification benchmarks, SPRiF attains the highest mean accuracy among the compared methods on three tasks and remains competitive on the other two. Ablations identify slow-fast state separation as the primary contributor, with consistent gains from spectral rotation and projective reset. A controlled spike-intervention task shows stable delay-period recall under reset stress, and impulse-response analysis reveals heterogeneous temporal kernels. These results support functional state decomposition as a practical neuron-level design principle for temporal SNNs.

1 Introduction

Spiking neural networks (SNNs) process temporal information through discrete spike events. They are used in speech recognition, physiological signal analysis, and event-based recognition. Unlike continuous-activation networks, their representational capacity depends on both inter-layer connectivity and neuron dynamics. These dynamics determine how neurons accumulate temporal context, decide when to fire, and update their states after each spike. When relevant information spans multiple timescales, from speech phonemes to long permuted sequences, neuron-level dynamics constrain the temporal patterns that an SNN can retain and exploit.

The leaky integrate-and-fire (LIF) neuron remains the dominant SNN unit because it is simple and compatible with surrogate-gradient training. Its extensions enrich neuronal

dynamics in several ways. Adaptive-threshold models introduce slow variables that modulate firing (???). Resonant neurons add damped subthreshold oscillations (???), and multi-timescale designs use heterogeneous neuronal or synaptic decays (???). Recent reset mechanisms also modify the post-spike update to reduce information loss (???). Despite these advances, the spike-generating variable is usually also the reset target. In standard LIF, one membrane potential accumulates history, determines spike timing, and absorbs reset. Each spike therefore perturbs its temporal trace. Oscillatory and adaptive variants retain this membrane-centred organisation, whereas reset innovations soften, adapt, or remove reset from the same spike-generating state.

This coupling raises a basic question: **must the state that triggers a spike also store temporal memory?** A spike indicates that a discharge-related variable has crossed a threshold and therefore requires correction. It need not alter every accumulated memory trace. Insulating a memory-bearing state from spike-triggered reset could preserve a richer input history without abandoning binary communication or enlarging the network.

We address this question with **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), which implements *functional state decomposition*. Each neuron contains a slow spectral state and a fast discharge state. The slow state integrates input through one real exponential mode and one two-dimensional damped rotational mode. It is never reset by spikes. The fast state receives a projection of the slow state, produces the membrane readout and binary spikes, and absorbs a learnable *projective reset* along $[1, \lambda]^T$. SPRiF is trained using standard backpropagation through time and surrogate gradients. Spike events therefore correct only the fast discharge state while the slow state evolves continuously.

Functional decomposition differs from conventional multi-timescale design. The latter assigns different decay rates to parallel channels, a form of *rate decomposition*, while coupling their roles to readout and reset. SPRiF instead assigns distinct functions. The constrained slow state stores temporal context, and the fast state implements readout and reset. The reset operator has no update path to the slow state. The distinction is therefore structural and testable at the update level, rather than a consequence of heterogeneous time constants alone.

We evaluate SPRiF on five temporal benchmarks spanning

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sequential image classification, physiological signal analysis, speech recognition, and event-based digit recognition. SPRiF attains the highest compared mean accuracy on S-MNIST, PS-MNIST, and QTDB. It remains close to the strongest compared results on GSC and SHD. Its end-to-end parameter count is lower than that of the strongest compared entry on four datasets and comparable on QTDB. Ablations on PS-MNIST, GSC, and QTDB isolate the contributions of rotation, state separation, and projective reset. A controlled spike-intervention experiment and impulse-response analyses further examine the proposed dynamics.

Contributions.

1. We propose *functional state decomposition* for spiking neurons, separating temporal memory storage, spike read-out, and post-spike reset into distinct state components. This principle is orthogonal to rate decomposition and single-state reset.
2. We introduce a *constrained spectral slow state* with real exponential decay and damped rotation modes, providing interpretable, structured temporal filtering beyond the single-exponential LIF trace.
3. We propose *projective fast-state reset*, a learnable directional reset mechanism that operates exclusively on the discharge state and leaves the memory state untouched.
4. We evaluate SPRiF on five temporal benchmarks, three mechanism ablations, a controlled spike-intervention experiment, and analyses of learned temporal kernels. Code will be released upon acceptance.

2 Related Work

2.1 Neuron Dynamics for Temporal Processing

Extensions of the leaky integrate-and-fire (LIF) neuron enrich temporal dynamics through gated integration, conductance-based modelling, adaptive thresholds, and improved surrogate-gradient training (????). Adaptive and multi-timescale models introduce slow variables, heterogeneous neuronal or synaptic decays, conductance channels, and learnable temporal gates (???????). These approaches provide *rate decomposition*: parallel components evolve at different timescales, but remain coupled to membrane read-out and reset.

Oscillatory and dendritic neurons offer complementary routes to temporal selectivity. Resonate-and-fire variants use damped oscillations (????), whereas compartmental and heterogeneous models introduce dendritic structure and dynamics (????). Structured state-space models have also been incorporated into spiking neurons (??). SPRiF differs from these alternatives along three axes. It assigns memory and discharge to distinct state roles, confines reset to the discharge state, and constrains memory dynamics to one real and one rotational spectral mode. Its contribution is therefore a structurally separated memory–reset interface, rather than heterogeneous rates or spatial compartments alone.

2.2 Reset and Functional State Separation

Existing reset mechanisms soften, adapt, or remove the post-spike update to reduce information loss and, in some cases,

enable parallel training (????). They still modify the spike-generating state itself, or remove reset from that state. SPRiF instead confines a learnable projective reset to the fast discharge state and leaves the slow spectral state untouched. Network-level temporal architectures, including temporal subnetworks and stochastic spike communication (??), are complementary to this neuron-level separation.

3 Method

3.1 Limitation of Conventional LIF Dynamics

We begin with the discrete-time dynamics of a standard leaky integrate-and-fire (LIF) neuron. For presynaptic activities $s_{i,t}$, the input current is

$$I_t = \sum_i w_i s_{i,t} + b, \quad (1)$$

where w_i is the synaptic weight of input i and b is a bias term. The membrane state is then updated by integrating this input, leaking its history, and incorporating the reset caused by the previous spike:

$$v_t = \beta v_{t-1} + I_t - z_{t-1} \theta, \quad (2)$$

$$z_t = H(v_t - \theta). \quad (3)$$

Here, $\beta \in (0, 1)$ is the leak factor, H is the Heaviside step function, and θ is the firing threshold. The same scalar state v_t therefore carries temporal context, determines spike timing, and receives the reset contribution from the preceding spike. Unrolling the recurrence over k steps gives

$$v_t = \beta^k v_{t-k} + \sum_{\ell=0}^{k-1} \beta^\ell (I_{t-\ell} - \theta z_{t-\ell-1}). \quad (4)$$

Both the external input history and the reset history are consequently filtered through the same leaky accumulation. Without reset, an input impulse contributes a single exponential trace proportional to β^k after k steps. With reset, each spike inserts a corrective term into that same trace. The reset operation is therefore not separable from the state that transports temporal information, which motivates a neuron in which memory storage and post-spike discharge occupy different state variables.

3.2 SPRiF Spiking Neuron Model

SPRiF is a two-path spiking neuron in which a spectrally structured slow state carries temporal context, while a resettable fast state converts that context into membrane voltage and binary spikes. For a layer input $\mathbf{s}_t$, the input current is

$$I_t = W_{\text{in}} \mathbf{s}_t + W_{\text{rec}} \mathbf{z}_{t-1} + b, \quad (5)$$

where the recurrent term is omitted in feed-forward layers. Each neuron maintains a three-dimensional slow state $\mathbf{x}_t \in \mathbb{R}^3$ and a two-dimensional fast state $\mathbf{u}_t \in \mathbb{R}^2$. The slow state stores temporal context, whereas the fast state produces the membrane readout, spike, and post-spike correction. With $\tilde{\mathbf{u}}_t$ denoting the pre-reset fast state, the complete discrete-time

dynamics of SPRiF are

$$\begin{cases} \mathbf{x}_t = A\mathbf{x}_{t-1} + B\mathbf{I}_t, \\ \tilde{u}_t^0 = \eta_0 u_{t-1}^0 + \kappa u_{t-1}^1 + (G\mathbf{x}_t)^0, \\ \tilde{u}_t^1 = \eta_1 u_{t-1}^1 + (G\mathbf{x}_t)^1, \\ v_t = \tilde{u}_t^0, \quad z_t = H(v_t - \theta), \\ \mathbf{u}_t = \tilde{\mathbf{u}}_t - z_t \theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}. \end{cases} \quad (6)$$

The computation therefore follows $I_t \rightarrow \mathbf{x}_t \rightarrow \tilde{\mathbf{u}}_t \rightarrow v_t \rightarrow z_t \rightarrow \mathbf{u}_t$. The first line of Eq. (??) updates the state that represents temporal context. The next two lines map this context into a two-dimensional discharge state, whose first coordinate is exposed as the membrane readout. Thresholding is applied only after this readout has been formed, and the final line performs the post-spike correction. Thus, the two paths meet through the learned projection G , but reset acts only on the fast path.

Equation (??) is written for one neuron, with the neuron index suppressed for readability. In a layer, the parameters $A, B, G, \eta_0, \eta_1, \kappa$, and λ_j are instantiated per neuron, while W_{in} and, when present, W_{rec} couple neurons through the input current. A SPRiF layer applies this update at every time step and emits $[z_1, \dots, z_T]$. Layers can be stacked, and recurrence is optional rather than intrinsic to the neuron definition. The following subsections characterize the memory and discharge paths separately.

3.3 Spectral Slow-Memory Dynamics

The slow-state update in Eq. (??) retains temporal context without direct spike-triggered correction. Rather than learning an unconstrained recurrent transition, SPRiF uses the smallest real-valued linear system containing one real decay mode and one complex-conjugate rotational pair, with

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho \cos \omega & -\rho \sin \omega \\ 0 & \rho \sin \omega & \rho \cos \omega \end{bmatrix}, \quad B = \begin{bmatrix} 1 - \alpha \\ 1 - \rho \\ 0 \end{bmatrix}. \quad (7)$$

The spectrum of A is

$$\text{eig}(A) = \{\alpha, \rho e^{i\omega}, \rho e^{-i\omega}\}, \quad (8)$$

The first coordinate is an exponential trace, $x_t^0 = \alpha x_{t-1}^0 + (1 - \alpha)I_t$, while (x_t^1, x_t^2) form a damped rotation driven through the first oscillatory coordinate. We parameterize

$$\alpha = \sigma(\alpha_{\text{raw}}), \quad \rho = \sigma(\rho_{\text{raw}}), \quad \omega = \pi \sigma(\omega_{\text{raw}}), \quad (9)$$

where σ is the logistic sigmoid. Hence, $\alpha, \rho \in (0, 1)$ and $\omega \in (0, \pi)$, so every eigenvalue of A lies strictly inside the unit disk. The parameter α controls real decay, ρ controls rotational damping, and ω controls rotational frequency. This parameterization provides a stable transition with an interpretable spectrum, rather than an unconstrained recurrent matrix whose temporal modes are not identifiable.

The resulting temporal basis is explicit. For a unit impulse entering the slow state, the response after k subsequent transitions is

$$A^k B = \begin{bmatrix} (1 - \alpha)\alpha^k \\ (1 - \rho)\rho^k \cos(k\omega) \\ (1 - \rho)\rho^k \sin(k\omega) \end{bmatrix}. \quad (10)$$

The real coordinate therefore supplies a monotone exponential trace, whereas the rotational pair supplies phase-shifted, damped oscillatory traces. Their combination gives each neuron a compact temporal basis with distinct decay and frequency controls while preserving a linear-time recurrence. Because this state is computed before spike generation and does not appear in the reset equation, the memory dynamics remain separate from post-spike correction.

3.4 Fast Discharge Dynamics and Projective Reset

The fast-state equations in Eq. (??) read out the slow state and convert it into a resettable discharge process. The coefficients $\eta_0, \eta_1 \in (0, 1)$ are learned fast leaks, κ couples the second fast coordinate to the first, and $G \in \mathbb{R}^{2 \times 3}$ projects slow dynamics into the discharge space. Equivalently, the autonomous fast transition is upper triangular, with diagonal entries η_0 and η_1 and off-diagonal coupling κ . The second fast coordinate can therefore modulate the membrane coordinate, whereas the membrane coordinate does not directly alter the second coordinate. This asymmetric coupling supplies additional discharge dynamics without placing a reset pathway in the slow state.

The first fast coordinate is exposed as the membrane potential $v_t = \tilde{u}_t^0$. The forward pass generates the binary spike $z_t = H(v_t - \theta)$ with the hard threshold, preserving the event-driven output interface of an SNN. During optimization, we replace only the derivative of the threshold with the multi-Gaussian surrogate used in recurrent SNN training (?); the forward dynamics remain those in Eq. (??).

After a spike, SPRiF applies a projective reset only to the fast state:

$$\mathbf{u}_t = \tilde{\mathbf{u}}_t - z_t \theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}, \quad (11)$$

where λ_j is learned independently for neuron j . Setting $\lambda_j = 0$ recovers scalar subtraction on the membrane coordinate. For nonzero λ_j , reset modifies both fast coordinates along a learned direction in the discharge space. Hence, λ_j controls the geometry of post-spike correction rather than the temporal memory transition. The reset direction is applied only after spike generation and introduces no additional update term for $\mathbf{x}_t$. All dynamics parameters, including $\alpha, \rho, \omega, G, \kappa, \eta$, and λ , are jointly optimized with the network weights through backpropagation through time.

3.5 Reset-Memory Separation Analysis

The preceding equations distinguish the direct effect of reset from the ordinary temporal propagation of spike signals. Relative to the pre-reset states, the reset-induced increments are

$$\Delta \mathbf{x}_t^{\text{reset}} = \mathbf{0}, \quad \Delta \mathbf{u}_t^{\text{reset}} = -z_t \theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}. \quad (12)$$

For LIF, the reset contribution $-z_{t-1}\theta$ enters the current scalar state in Eq. (??); as shown by Eq. (??), it is accumulated through the same recurrence as the input history. In SPRiF, the reset-induced correction is confined to $\mathbf{u}_t$, while the slow state $\mathbf{x}_t$ is unchanged at the same time step. This separation preserves a resettable spike-generation pathway without directly rewriting the spectral memory state.

The slow-state recurrence makes this distinction explicit. Iterating the first line of Eq. (??) yields

$$\mathbf{x}_t = A^t \mathbf{x}_0 + \sum_{\tau=1}^t A^{t-\tau} B I_\tau. \quad (13)$$

No local reset term appears in Eq. (??); equivalently, holding I_t fixed gives $\partial \mathbf{x}_t / \partial z_t = \mathbf{0}$ for the reset operation. This statement does not imply that spikes are ignored by the network. In recurrent layers, $\mathbf{z}_t$ enters the subsequent input through $W_{\text{rec}} \mathbf{z}_t$ in Eq. (??), and therefore affects $\mathbf{x}_{t+1}$ through the ordinary input pathway. SPRiF thus separates two effects that are coupled in conventional LIF: spikes communicate information to subsequent network updates, whereas local post-spike reset does not directly overwrite the state that stores temporal context.

4 Experiments

4.1 Benchmarks and Protocol

We evaluate SPRiF on five temporal classification benchmarks spanning sequential images (S-MNIST and PS-MNIST), physiological signals (QTDB), frame-based speech (GSC), and event-driven speech (SHD). Accuracy is the evaluation metric throughout. These tasks differ substantially in input encoding, sequence length, and temporal structure, allowing us to test whether the proposed neuron dynamics transfer beyond a single temporal modality.

We compare SPRiF with published recurrent SNNs that use gated, adaptive, dendritic, resonate-and-fire, and other temporal neuron dynamics. The architectures and optimization protocols are adapted to each benchmark, whereas the reported SPRiF variants retain the same neuron definition. Complete dataset preprocessing, task-specific architectures, optimization settings, initialization procedures, and reproducibility details are provided in the technical appendix. Because the source studies use task-specific protocols, Table ?? is interpreted as a descriptive cross-paper comparison rather than a significance test.

4.2 Main Results

Table ?? summarizes the benchmark comparison. SPRiF obtains the highest reported accuracy among the listed methods on S-MNIST, PS-MNIST, and QTDB, while remaining competitive on GSC and SHD. The comparison covers five distinct temporal input regimes, indicating that the proposed state decomposition is not restricted to a single encoding modality. Parameter counts are reported alongside accuracy to provide architectural context; they should not be interpreted as controlled cross-paper efficiency comparisons because the published systems differ in architecture and training protocol.

4.3 Mechanism Ablations

We isolated three design choices on PS-MNIST, GSC, and QTDB. Ablation A set $\omega = 0$, removing rotational coupling while retaining the real decay path. Ablation B merged memory and discharge into a single three-dimensional state, whose first coordinate served as both membrane and reset

target. Ablation C fixed $\lambda = 0$, reducing Eq. (??) to scalar reset. All variants otherwise retained the task-specific training configuration.

Table ?? shows two consistent patterns. Merging the states reduced accuracy by 2.38–5.04 points. Thus, the benefit of state separation extends beyond increased state dimensionality: the same three-dimensional capacity without the slow-fast split performed substantially worse. All variants retained matched task-specific training configurations. Rotation contributed 0.65–3.53 points, with the largest effect on the long permuted sequence. Projective reset provided a smaller but consistent gain of 0.33–1.03 points. Across these settings, state separation was the largest ablated component, whereas spectral rotation and a learnable reset direction further refined temporal processing.

4.4 Controlled Spike Intervention

To test whether slow-fast separation protects memory under reset stress, we designed a Spike-Intervention Delayed Match-to-Sample (SI-DMS) task. A cue was followed by a delay, during which selected hidden neurons were forced to spike at controlled times. A match/non-match probe then tested recall. At each intervention timestep, 10% of hidden neurons were forced to spike before thresholding, so that the standard reset mechanism was applied. We evaluated $K \in \{0, 1, 2, 4, 8\}$ intervention times. The maximum-stress setting, $K = 8$, was not used during training. If reset reaches the memory-bearing state, these interventions should impair delay-period recall. If the slow state is structurally insulated, recall should remain more stable.

Table ?? reports clean ($K = 0$) and maximum-stress ($K = 8$) accuracy averaged over three seeds. The fully separated variant, `sprif_full`, reached 92.8% clean accuracy and lost 0.7 points under intervention stress. Removing the projective reset direction ($\lambda=0$; `sprif_lambda0`) produced a comparable 0.6-point drop. This pattern indicates that structural separation, rather than λ alone, was important under this protocol. The merged variant, `sprif_merged`, performed near chance in both conditions (51.7% clean and 50.4% under maximum stress). Among the external baselines, BRF decreased from 70.4% to 64.9%, whereas LIF and AS-RNN remained near chance throughout. Under matched SI-DMS controls, these results support the claim that functional state decomposition protects delay-period memory from spike-triggered disruption. Figure ?? shows the delay-by-intervention grid. Complete grids and integrity audits are provided in the appendix.

4.5 Dynamical Analysis

The update equations ensure that reset does not directly modify $\mathbf{x}_t$. SI-DMS tests the functional consequence of this design under controlled interventions. Figure ??a shows the resulting state dynamics around recorded spikes. Figure ??b shows a slow-state oscillation across probe windows in a cue-delay task. It illustrates continuous slow-state evolution, rather than providing an event-level reset test. Figure ??c–e examines the learned projective reset directions.

Projective reset did not collapse to a shared direction. Across QTDB, GSC, and PS-MNIST, the learned λ values

Table 1: Comparison across five temporal benchmarks. “Rec” denotes a recurrent network, and parameters are reported in millions (M). SPRiF entries (bold) are mean $\pm$ std over 5 seeds.

Datasets	Method	Network	Parameters (M)	Accuracy (%)
S-MNIST	GLIF _{NeurIPS, 2022} (?)	Rec	0.15	96.64
	ASRNN _{Nat. Mach. Intell., 2021} (?)	Rec	0.15	98.70
	DH-SNN _{Nat. Commun., 2024} (?)	Rec	0.08	98.90
	BRF _{ICLR, 2024} (?)	Rec	0.069	99.10
	TC-LIF _{AAAI, 2024} (?)	Rec	0.15	99.20
	SPRiF (ours)	Rec	0.067	99.28 $\pm$ 0.06
PS-MNIST	GLIF _{NeurIPS, 2022} (?)	Rec	0.15	90.47
	ASRNN _{Nat. Mach. Intell., 2021} (?)	Rec	0.15	94.30
	DH-SNN _{Nat. Commun., 2024} (?)	Rec	0.08	94.52
	BRF _{ICLR, 2024} (?)	Rec	0.069	95.20
	TC-LIF _{AAAI, 2024} (?)	Rec	0.15	95.36
	SPRiF (ours)	Rec	0.067	95.86 $\pm$ 0.14
QTDB	ASRNN _{Nat. Mach. Intell., 2021} (?)	Rec	0.0018	85.90
	DH-SNN _{Nat. Commun., 2024} (?)	Rec	0.00178	86.35
	BRF _{ICLR, 2024} (?)	Rec	0.00173	87.00
	SE-adLIF _{Nat. Commun., 2025} (?)	Rec	0.0018	86.88
	SPRiF (ours)	Rec	0.00177	88.43 $\pm$ 0.36
GSC	SNN-SFA _{eLife, 2021} (?)	Rec	4.19	91.21
	ASRNN _{Nat. Mach. Intell., 2021} (?)	Rec	0.31	92.10
	RadLIF _{Front. Neurosci., 2022} (?)	Rec	1.20	94.51
	DH-SNN _{Nat. Commun., 2024} (?)	Rec	0.13	93.80
	TC-LIF _{AAAI, 2024} (?)	Rec	0.20	94.84
	SPRiF (ours)	Rec	0.13	94.55 $\pm$ 0.31
SHD	Heterogeneous SNN _{Nat. Commun., 2021} (?)	Rec	0.11	82.50
	ASRNN _{Nat. Mach. Intell., 2021} (?)	Rec	0.14	90.40
	DH-SNN _{Nat. Commun., 2024} (?)	Rec	0.05	91.34
	BRF _{ICLR, 2024} (?)	Rec	0.11	91.70
	TC-LIF _{AAAI, 2024} (?)	Rec	0.14	88.91
	MPS-SNN _{ICLR, 2025} (?)	Rec	0.39	91.19
	DGN _{ICLR, 2026} (?)	Rec	0.22	87.78
	SPRiF (ours)	Rec	0.05	91.52 $\pm$ 0.46

Table 2: Mechanism ablations (accuracy, %). Each component contributes on all three datasets, with merging memory and discharge causing the largest degradation.

Dataset	Full	$\omega=0$	Merged	$\lambda=0$
PS-MNIST	95.82	92.29	93.44	95.32
GSC	94.55	93.83	90.05	94.22
QTDB	88.43	87.78	83.39	87.40

spanned both signs (Fig. ??c–d). Their correlations with firing rate, α , and ω were weak, with $|r| \leq 0.191$ (Fig. ??e). Together with the $\lambda = 0$ ablation loss, these observations support diverse reset directions in the fast state. They do not assign a universal biological meaning to either sign.

We next injected a unit impulse into trained neurons and recorded 100 steps of intrinsic slow-state evolution. Figure ?? visualizes the resulting temporal basis. α -stratified examples span rapidly decaying and persistent real traces, whereas the oscillatory coordinates vary in sign and frequency. All-neuron spectra show broad within-task variation in effective low-pass behaviour. In selected slow-decay examples, the SPRiF real modes had derived time constants of approxi-

Table 3: SI-DMS results (3-seed average). Clean and maximum-stress accuracies correspond to $K = 0$ and $K = 8$, respectively; “Drop” is the stress-induced accuracy loss. External baselines are marked (ext.). The separated SPRiF variants maintain high accuracy under intervention.

Model	Clean Acc.	Stress Acc.	Drop
sprif_full	0.928	0.921	0.007
sprif_lambda0	0.935	0.929	0.006
sprif_merged	0.517	0.504	0.013
LIF (ext.)	0.500	0.502	−0.002
ASRNN (ext.)	0.520	0.517	0.003
BRF (ext.)	0.704	0.649	0.055

mately 272 steps on GSC and 138 on QTDB. The closest available trained ASRNN membrane constants were 34 and 46 steps. This comparison is limited to GSC and QTDB, the only two tasks with trained ASRNN kernels in the saved analysis data. Additional impulse-response and spectral analyses are provided in the appendix.

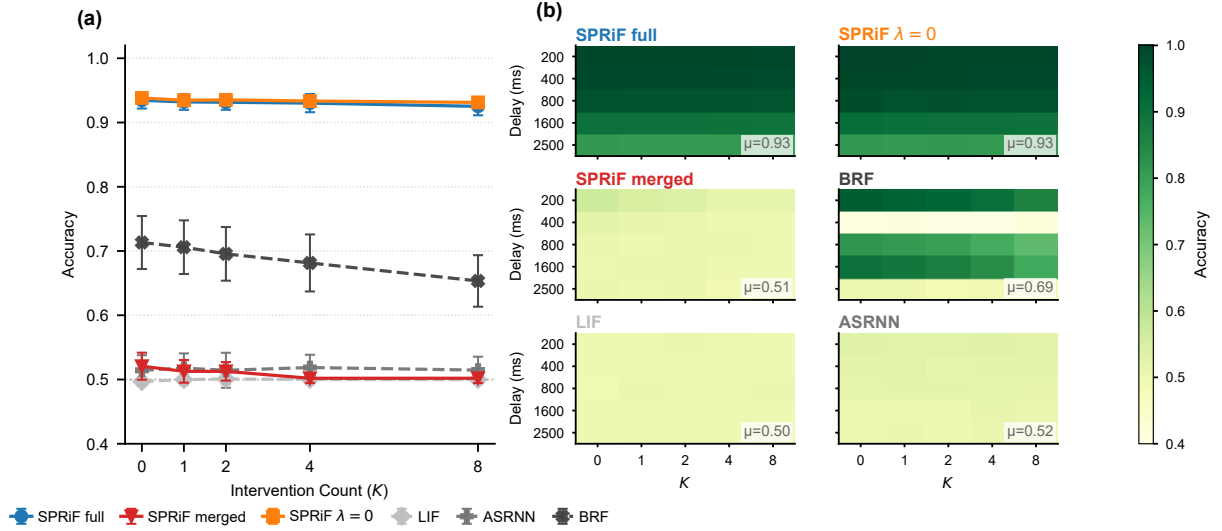


Figure 1: Controlled spike-intervention results (SI-DMS). (a) Accuracy versus intervention count K for the three SPRiF mechanism variants (solid, saturated) and three external baselines (dashed, gray); values are 3-seed means $\pm$ seed standard deviations. (b) Per-model delay-by- K accuracy grids (3-seed mean) for all six models with a shared colorbar; μ denotes the per-model mean across the grid.

5 Conclusion

SPRiF is built on a simple principle: the state that stores temporal context need not be reset after a spike. A constrained spectral slow state supplies real and rotational memory modes. A fast discharge state produces binary spikes and receives a learnable projective reset. Across five temporal benchmarks, this design attained the highest mean accuracy in our comparison on three tasks and remained competitive on the other two with modest network size. Ablations identified slow-fast separation as the largest contributor, with additional gains from rotation and projective reset. Controlled spike-intervention and impulse-response analyses provided complementary evidence for the proposed mechanism. These findings support functional state decomposition as a reusable neuron-level design direction. Larger-scale evaluation and neuromorphic-hardware deployment remain open.

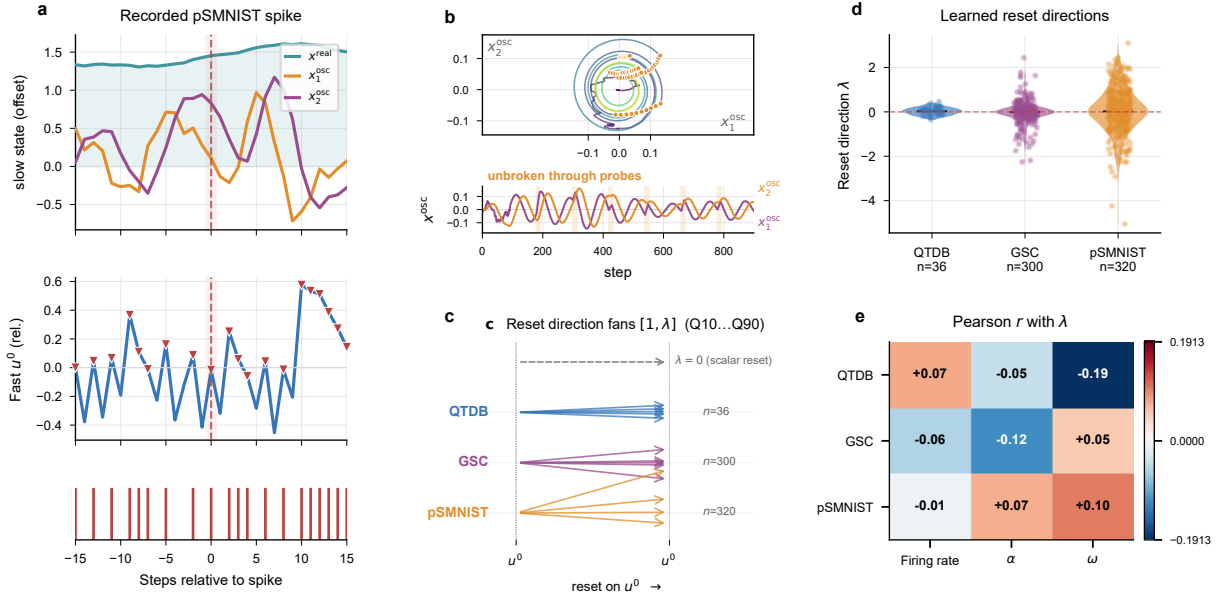


Figure 2: Functional decomposition and learned projective reset. (a) Slow and fast states of a trained PS-MNIST neuron around a recorded spike; slow coordinates evolve continuously while the fast membrane trace changes abruptly. (b) Oscillatory slow-state trajectory in a cue-delay task: the 2D phase plane (top) marks probe times, and the x_1^{osc} and x_2^{osc} time series (bottom) traverse the orange probe windows without a discontinuity, illustrating continuous slow-state evolution rather than an event-level reset test. (c) Representative learned reset directions $[1, \lambda]$ from task-wise quantiles. (d) Full distributions of λ . (e) Pearson correlations between λ and firing rate, α , and ω .

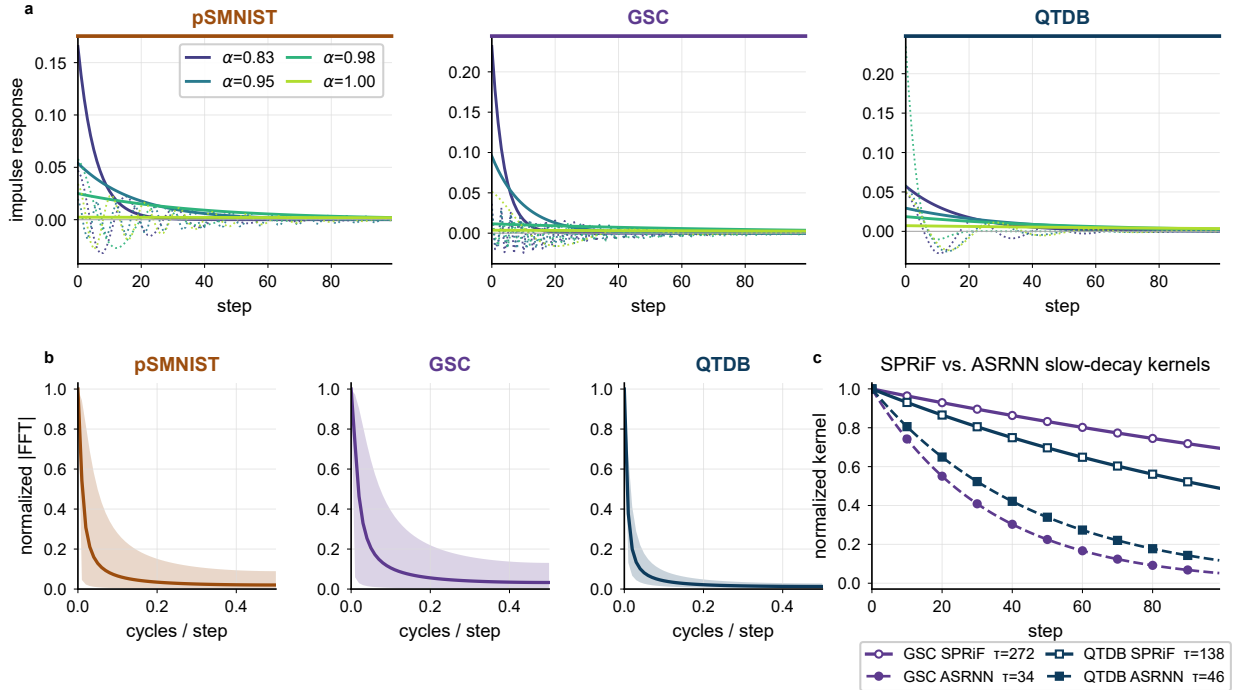


Figure 3: Learned temporal kernels from impulse-response analysis. (a) Real (solid) and oscillatory (dotted) slow-state coordinates of four α -quantile SPRiF neurons per task over 100 post-impulse timesteps; traces span rapidly decaying to persistent regimes. (b) Median normalized real-coordinate amplitude spectrum with the 10–90% range aggregated over all neurons, per task. (c) Representative slow-memory SPRiF kernels (real coordinate) and the closest available trained ASRNN membrane kernels on GSC and QTDB; derived time constants τ (timesteps) are shown.